Problem Set 2

The Constrained Optimisation Problem

EC306: Intermediate Microeconomics

Instructions. Work through these in class. Show all working and interpret your numerical answers in words. Worked solutions follow at the end.

Question 1 — Lagrangian with three goods

A consumer has $u(x, y, z) = x^a y^b z^c$ with $a, b, c > 0$, facing prices p_x, p_y, p_z and income m.

(a) Write the Lagrangian and the four first-order conditions.

(b) Solve for the Marshallian demands $x^\star, y^\star, z^\star$. Show each expenditure share equals the corresponding normalised exponent.

(c) Derive an expression for the multiplier $\lambda^\star$ in terms of prices, income, and the exponents.

(d) State precisely what $\lambda^\star$ measures and verify, by differentiating the indirect utility $u(x^\star, y^\star, z^\star)$ with respect to m, that $\partial u^\star / \partial m = \lambda^\star$. In one sentence, explain why this object is called a "shadow price".

Question 2 — Interior vs corner

A consumer has $u(x, y) = x + 2\sqrt{y}$ (quasilinear in x), income m, prices p_x, p_y.

(a) Derive the interior demand for y and show it depends only on the price ratio, not on m.

(b) Derive the demand for x and state the exact income threshold $\underline{m}$ below which the consumer is at a corner with $x = 0$.

(c) Sketch (by hand, include in your submission) the Engel curve for y — i.e. $y^\star$ as a function of m — and explain its kink in terms of the income effect on y.

Question 3 — Cash vs in-kind, with numbers

A household has $u(f, c) = f^{0.4} c^{0.6}$, where f is food and c is a composite of all else, $p_f = p_c = 1$, income $m = 100$.

(a) Find the baseline optimum (f_0, c_0) and the utility u_0.

(b) The government gives a food voucher worth $T = 50$ usable only on food. Write the new (kinked) budget constraint and find the new optimum. Does the household spend more than 50 on food?

(c) Now suppose instead the government gives £50 cash. Find that optimum and its utility.

(d) Compare the two utilities. Are they equal or does cash strictly dominate *here*? Explain in terms of whether the voucher constraint binds, and state the general condition (in words) under which a voucher is welfare-equivalent to cash.

Question 4 — Your own budget, your own numbers

(a) Pick two goods you personally buy regularly (name them). Find their *current* unit prices and cite sources. Choose a sensible weekly budget m and state it.

(b) Assuming Cobb–Douglas preferences, choose exponents (a, b) that reflect roughly how you split spending between them, and justify the split in one sentence.

(c) Compute the implied optimal quantities. Then state how the optimal bundle would change if the price of the first good rose 20%, giving the new quantities.

(d) Your Cobb–Douglas model predicts the *share* on each good is unchanged by that price rise. Is that realistic for the specific goods you chose? Argue for or against in 3–4 sentences, naming what feature of the real goods the model misses.

Solutions

Question 1

(a) $\mathcal{L} = x^a y^b z^c + \lambda\big(m - p_x x - p_y y - p_z z\big)$. FOCs:

$$a\,x^{a-1}y^b z^c = \lambda p_x, \quad b\,x^a y^{b-1} z^c = \lambda p_y, \quad c\,x^a y^b z^{c-1} = \lambda p_z, \quad p_x x + p_y y + p_z z = m.$$

(b) Multiply each FOC by the relevant quantity: $a\,u = \lambda p_x x$, $b\,u = \lambda p_y y$, $c\,u = \lambda p_z z$, so $p_x x = \frac{a}{\lambda}u$, etc. Summing and using the budget, $m = \frac{a+b+c}{\lambda}u$. Writing $s = a+b+c$,

$$x^\star = \frac{a}{s}\frac{m}{p_x}, \quad y^\star = \frac{b}{s}\frac{m}{p_y}, \quad z^\star = \frac{c}{s}\frac{m}{p_z}.$$

Expenditure shares: $p_x x^\star/m = a/s$, and likewise b/s, c/s — the normalised exponents.

(c) From $m = su/\lambda$, $\lambda^\star = \dfrac{s\,u^\star}{m}$, where $u^\star = \big(\frac{a}{s}\big)^a\big(\frac{b}{s}\big)^b\big(\frac{c}{s}\big)^c \dfrac{m^s}{p_x^a p_y^b p_z^c}$, giving $\lambda^\star = s\big(\frac{a}{s}\big)^a\big(\frac{b}{s}\big)^b\big(\frac{c}{s}\big)^c \dfrac{m^{s-1}}{p_x^a p_y^b p_z^c}$.

(d) $\lambda^\star$ is the *marginal utility of income*. Since $u^\star \propto m^s$, $\partial u^\star/\partial m = s\,u^\star/m = \lambda^\star$. It is a "shadow price" because it is the utility value of relaxing the budget constraint by one unit (one extra £of income).

Question 2

(a) $\mathrm{MU}_x = 1$, $\mathrm{MU}_y = 1/\sqrt{y}$. Tangency $\mathrm{MU}_y/\mathrm{MU}_x = p_y/p_x \Rightarrow 1/\sqrt{y} = p_y/p_x \Rightarrow \sqrt{y} = p_x/p_y$, so $y^\star = (p_x/p_y)^2$ — depends only on the price ratio, not on m.

(b) Budget: $x^\star = \dfrac{m - p_y y^\star}{p_x} = \dfrac{m}{p_x} - \dfrac{p_x}{p_y}$. This is non-negative only if $m \geq \underline{m} = \dfrac{p_x^2}{p_y}$. Below $\underline{m}$ the consumer is at the corner $x = 0$ and spends all income on y.

(c) Engel curve for y: for $m \geq \underline{m}$, $y^\star = (p_x/p_y)^2$ is *flat* (all extra income goes to x — the quasilinear good has zero income effect on y). For $m < \underline{m}$, $x = 0$ and $y^\star = m/p_y$ rises linearly. The kink at $m = \underline{m} = p_x^2/p_y$ marks the switch from a positive income effect on y (corner) to a zero income effect on y (interior).

Question 3

(a) Cobb–Douglas with $p_f = p_c = 1$, $m = 100$: $f_0 = 0.4(100) = 40$, $c_0 = 0.6(100) = 60$. $u_0 = 40^{0.4}60^{0.6} \approx 51.0$.

(b) Voucher $T = 50$, food-only: budget is $f + c = 150$ subject to $c \leq 100$ (equivalently $f \geq 50$). The unconstrained optimum at income 150 is $f = 0.4(150) = 60$, $c = 0.6(150) = 90$, which satisfies $f \geq 50$ and $c \leq 100$, so the kink does *not* bind. New optimum $(f_1, c_1) = (60, 90)$; the household spends $60 > 50$ on food (it tops the voucher up with £10 of its own cash).

(c) Cash £50: income 150, no restriction $\Rightarrow f = 60$, $c = 90$ — identical to (b). $u_1 = 60^{0.4}90^{0.6} \approx 76.5$.

(d) The two utilities are **equal here**: the voucher does not bind because desired food spending (60) already exceeds the voucher (50). General rule: an in-kind voucher is welfare-equivalent to cash *iff* its value does not exceed what the consumer would freely spend on that good anyway (the constraint is slack). If the voucher exceeded desired food spending it would bind and cash would strictly dominate.

Question 4 (model answer / marking notes)

Data/choice-dependent. A strong answer: **(a)** names two regularly-bought goods with current cited unit prices and a stated weekly budget m. **(b)** picks exponents (a, b) summing to 1 that match the rough spending split, with a one-line justification. **(c)** computes $x^\star = a\,m/p_x$,

$y^\star = b\,m/p_y$; after a 20% rise in p_x, $x^\star$ falls by the factor $1/1.2$ ($x'^\star = a\,m/(1.2p_x)$) while $y^\star$ is unchanged. **(d)** the Cobb–Douglas prediction of *constant expenditure shares* (unit own-price elasticity, zero cross-price effect) is unrealistic for goods that are close substitutes or strong complements, or where habit/stockpiling matters — name the specific feature the chosen goods exhibit.